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Edexcel GCSE Geography B (1GB0) — Component 3: People and Environment Issues · Topic 7: People and the Biosphere

3.1 People and the Biosphere

1. Biomes and Global Climate Zones

A **biome** is a very large scale ecosystem, defined by its climate and the plant and animal communities that climate supports. Biomes are not scattered at random: their distribution is controlled almost entirely by the **global circulation of the atmosphere**, which fixes where air rises (low pressure, wet) and where it sinks (high pressure, dry). Because that circulation is arranged in belts either side of the equator, the world's biomes also form broad latitudinal belts.

> **Biome:** A global scale ecosystem in which the climate, soils, vegetation and animal life are closely interdependent — for example tropical rainforest, hot semi-arid grassland (savanna), tundra or temperate deciduous forest.

How Global Circulation Fixes Biome Distribution

Latitude	Air movement	Pressure and rainfall	Resulting biome
0° (equator)	Rising air at the ITCZ	Low pressure; rain all year, over 2,000 mm	Tropical rainforest
5–20° N and S	ITCZ migrates north and south with the overhead sun	Alternating low and high pressure; a wet season and a long dry season	Hot semi-arid grassland (savanna)
20–30° N and S	Descending limb of the Hadley cell	Permanent high pressure; under 250 mm	Hot desert

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40–60° N	Rising air where polar and tropical air meet	Low pressure; rain spread through the year	Temperate deciduous forest
60–70° N	Cold, dry, sinking air	High pressure; low precipitation, mostly snow	Tundra

Edexcel B Topic 7 asks you to understand the **distribution of the world's major biomes** and the factors that control it, the **biosphere as a life-support system** providing goods and services, and the growing **demand for resources** — including the contrasting views of Malthus and Boserup.

Exam Tip — Describing Distribution: When asked to describe where a biome is found, always name the latitude band, name two or three specific regions or countries, and link the pattern to pressure. "Savanna is found roughly between 5° and 20° north and south of the equator — across the Sahel, East Africa, northern Australia and the Brazilian cerrado — where the ITCZ brings rain for part of the year and high pressure dominates for the rest." That single sentence scores far more than "it is found in hot places".

2. Factors Affecting the Distribution of Biomes

Climate is the main control on where biomes are found, but local factors also matter:

Factor	Effect
Temperature	Decreases with latitude and altitude, controlling the length of the growing season
Precipitation	Determines whether forest, grassland or desert can grow
Sunshine hours	Affect photosynthesis and productivity
Altitude	Temperature falls about 6.5°C per 1,000 m, so mountains have bands of vegetation (e.g. Mount Kilimanjaro from savanna to ice)
Rock type and soils	Influence drainage and nutrients — e.g. chalk grassland, peat bogs

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Drainage	Waterlogged areas form wetlands and marshes

3. Energy Flows and Food Webs

Nutrient cycling moves matter round and round an ecosystem. **Energy does not cycle** — it enters as sunlight, passes along a one-way chain, and is lost as heat. Understanding that difference is the point of this part of the specification.

Producer: A green plant that converts sunlight into chemical energy by photosynthesis. Producers form the base of every food web.
Consumer: An animal that obtains energy by eating other organisms — primary consumers eat plants, secondary consumers eat primary consumers, tertiary consumers eat secondary consumers.
Decomposer: A bacterium, fungus or detritivore that breaks down dead material and returns nutrients to the soil.

Trophic Levels

Each step in a food chain is a **trophic level**. Only about **10% of the energy** at one level is passed on to the next — the rest is used in respiration, movement and body heat, or is lost in waste. This is why food chains rarely have more than four or five links, and why there are always far fewer top predators than producers.

Trophic level	Tropical rainforest	Hot semi-arid grassland
Producers	Canopy trees, lianas, epiphytes, forest-floor ferns	Red oat grass, elephant grass, acacia, baobab
Primary consumers	Leaf-cutter ants, sloths, howler monkeys, tapir	Wildebeest, zebra, gazelle, elephant, termites
Secondary consumers	Poison dart frogs, tree snakes, army ants	Jackal, serval, secretary bird

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Tertiary consumers	Jaguar, harpy eagle, anaconda	Lion, cheetah, spotted hyena
Decomposers	Fungi, bacteria, termites — working within days in the constant warmth and damp	Termites and dung beetles in the wet season; decomposition almost stops in the dry season

Why the Two Biomes Differ

- **Rainforest — energy concentrated at the base.** Constant warmth, light and water mean net primary productivity is the highest of any land biome. Most of the biomass sits in the standing trees, and most consumers are small-bodied specialists living in the canopy.
- **Savanna — energy concentrated in large herbivores.** Productivity is lower and strongly seasonal, but grass is far easier to eat than wood, so an unusually large share of the plant growth is converted into herbivore biomass. This is why the savanna, not the rainforest, carries the world's great herds.
- **Decomposition speed.** In the rainforest, leaf litter is broken down in weeks and nutrients are taken straight back up by shallow roots. In the savanna, dead grass lies dry and unrotted for months, which is why fire, rather than decomposition, is often what recycles it.

Biodiversity and Food Web Stability

A food web with many species at each level is more **resilient**: if one prey species declines, predators can switch to another, and the web holds. A simplified web — for example one created when farming replaces natural vegetation with a single crop — has no such alternatives, so the loss of one species can collapse several others. This is the mechanism behind the damage described in 3.2.

Exam Tip — Drawing and Reading Food Webs: Arrows in a food web always point in the direction that *energy flows*, i.e. from the organism being eaten to the organism eating it. A common and costly error is to draw them the other way. If asked to predict the effect of removing a species, trace every arrow that touches it and describe both the level above (loses a food source) and the level below (population rises).

4. The Biosphere as a Life-Support System

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Biosphere — the living layer of the Earth, including all plants, animals and other organisms on land, in water and in the air. **Ecosystem services** — the benefits people get from the biosphere.

Type	Examples
Goods (provisioning)	Food, fresh water, timber, fuel wood, medicines (about a quarter of modern medicines are derived from plants), fibres such as cotton
Regulating services	Regulating the composition of the atmosphere (photosynthesis produces oxygen and stores carbon), regulating climate, purifying water, controlling floods and soil erosion, pollination
Supporting services	Nutrient cycling, soil formation, maintaining biodiversity
Cultural services	Recreation, tourism, spiritual and aesthetic value

- **Indigenous peoples** depend directly on the biosphere for food, shelter, medicine and their way of life — e.g. the Yanomami in the Amazon and the Sámi reindeer herders in the Arctic.
- The biosphere is increasingly exploited for **commercial** purposes: energy (oil, biofuels), minerals, timber, palm oil and soya for global markets.

5. Growing Demand for Resources

- Population growth, rising wealth and urbanisation are increasing demand for **food, water and energy**. Global energy demand is expected to keep rising, mostly in emerging economies; water demand could exceed supply by 40% by 2030 under current trends; food production may need to rise by around 50% by 2050.
- This puts pressure on ecosystems through deforestation, overfishing, over-abstraction of water and pollution.

6. Malthus vs Boserup

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	Thomas Malthus (1798) — pessimistic	Ester Boserup (1965) — optimistic
Main idea	Population grows geometrically (1, 2, 4, 8…) but food supply grows arithmetically (1, 2, 3, 4…), so population will outstrip food	"Necessity is the mother of invention" — as population grows, people develop new technology to increase food production
Outcome	"Positive checks" — famine, disease and war — reduce population; or "preventive checks" such as later marriage	Food supply keeps pace with population through innovation, e.g. irrigation, fertilisers, GM crops
Evidence for	Famines (Ethiopia 1984, Somalia 2011); environmental degradation; water shortages	Green Revolution — world food production has more than kept pace with population since 1960
Evidence against	Global population has grown eightfold since 1798 without the predicted collapse	Around 700 million people are still undernourished; technology has environmental costs

Exam Tip: For "Assess which theory best explains…" questions, note that both have some truth: Boserup explains global trends, while Malthus's warnings about limits may apply to particular regions or to environmental resources such as fresh water and a stable climate.

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